Question I is about the processing of a starter for fertilizers

A naturally occuring mineral **A** decomposes at 1000°C. 44% (m/m) of the starting material are converted to a gaseous product **B** (density ρ at STP = 1.94 g/L). The remaining solid **D** reacts with coke in an electric-arc furnace to give another solid **E** and a gas **G**, which is isoelectronical to cyanide.

The product **E** hydrolyses easily to form a combustible gas **J**. A second product of a basic compound **K** is formed. It is also possible to convert the substance **E** with nitrogen at 1000 – 1100°C, which is a strong exothermic reaction. Therefore the temperature of this reaction will be reduced in the further processing. Aside of elemental carbon one gets a reactive solid **L**, which contains 15% (m/m) of carbon and 35% (m/m) of nitrogen as well as one metallic ion per formula unit.

Hydrolysis of **L** yields compound **K** and an intermediate **M**, which consecutively reacts with CO_2 and water forming compound **A** and a molecular solid **Q**:

$$2\,L + 2\,H_2O \rightarrow K + M$$

$$M + H_2O + CO_2 \rightarrow A + 2\,Q$$

There are two constitutional isomers of **Q**, one of them being symmetrical the other one unsymmetrical. Only the molecule without a symmetry center exists and shows in an IR-spectrum an absorption band between 2220 and 2260 cm^{-1}.

Substance **Q** is used for the processing of fertilizers. Its hydrolysis leads to compound **R**. **R** itself may be used as a fertilizer. In the year 1828 Friedrich Wöhler had synthesized this substance by heating the salt **T**. Hydrolysis of **R** gives two gaseous substances **B** and **X**. The latter has a pungent scent.

Another possibility to prepare compound **Q** is the reaction between a pseudo-interhalogen **Z** and a compound **X**. As a side product of this synthesis hydrogen chloride is formed.

a) Write down the total molecular formulae of the compounds **A** to **Z**!

b) Draw the two possible Lewis-structures of compound **Q** and assign clearly the existing substance by encircling the proper formula!

c) Write balanced equations for all the reactions.

d) Compound **E** has NaCl-structure, where the cations occupy the positions of the chloride-ions in the NaCl-lattice, and the anions of **E** occupy the octahedral vacancies. The density of **E** equals 2.22 g/mL. Calculate the distance between two cations of **E** in the unit pm.

Question II is Extraction of penicillin

Penicillin V is one of the main products of Sandoz. An important processing step is the extraction of penicillin from the fermentation mixture. This extraction is carried out using butyl acetate. It has a distribution coefficient of $K_D = 48.00$. Penicillin (to be more accurate the monoprotonic penicillanic acid HP) may undergo a protolysis in aqueous solution having a pK_A-value of 2.75. Therefore the distribution ratio D depends on the pH of the solution.

a) Derive a relationship between the distribution ratio D and the H+-concentration.

b) Calculate the distribution ratio D for the pH-values 1.00, 2.00, 3.00, 4.00 und 5.00.

c) Draw a diagram showing the correlation between D and the pH-region 1 to 5.

d) The concentration of penicillin in the aqueous solution of the fermentation tank (250 m^3) equals 3.00 kg/m^3, the pH being 2.00. Calculate the volume of butyl acetate necessary, if 99.0% of the penicillin should be extracted in one step.

e) Calculate the total volume of butyl acetate, if 99.0% of the penicillin are extracted in five steps using equal volumes in each of them.

f) To establish the necessary pH of 2.00 one uses concentrated sulfuric acid.

How many moles of H_2SO_4 must be added to one liter of water to reach this value? You may neglect the autoprotolysis of water and you may also assume that at first all the H_2SO_4 is transformed to give HSO_4^- (H_2SO_4: $pK_{A1} = -3$, $pK_{A2} = 1.92$).

g) The sulfuric acid used to acidify the fermentation mixture (250 m^3) has a concentration of 96% (m/m)(d = 1830 g/l). Which volume of sulfuric acid is necessary to establish pH = 2.00 according to the calculation in f). You may neglect all possibe side reactions with other compounds in the fermentation tank.

Question III is abput Unknown

Metal containing salt **A** can be prepared in a simple exchange reaction when the cold, saturated aqueous solutions of the two distinctly colored compounds, **B** and **C** are mixed in stoichiometric ratio. 10.00 g **B** in solution mixed with the solution of 12.86 g **C**, and immediately cooled to 2 °C will give 4.90 g solid **A**. The yield of A is 32.6 %.

To determine the composition of **A**, first iodometry is performed. A known mass of **A** is added to a titration flask, acidified with sulfuric acid, then KI is added in excess and there is a precipitate formation. After a few minutes, sodium citrate solution is added until the solution is free of precipitate. The citrate ions form a strong complex with a metal ion present in the mixture, reversing completely the reaction leading to the precipitate. The resulting mixture is titrated with sodium thiosulfate solution (Titration I via reaction 1). To the blue titrated solution another portion of sulfuric acid (significantly more than the initial amount) is added to protonate the citrate. The previous solid precipitate is formed once again [reaction 2]. The mixture is titrated with the same thiosulfate solution (titration II via reaction 1).

The average volumes for 100.0 mg **A** with $5.000 \cdot 10^{-2}$ M thiosulfate are 54.12 mL in Titration I and 5.41 mL in Titration II.

When the aqueous solution of **A** is heated, the blue precipitate **D** can be observed [reaction 3]. Compound **E** can be crystallized from the liquid above **D**. The low temperature during the synthesis of **A** is important to avoid contamination by **E**.

3.1 Write balanced equations for reactions [1] and [2].

The thermal decomposition of **A** was studied in detail. When pure **A** is slowly heated, already at around 75 °C it explodes. When the compound is dispersed in aluminium oxide and the mixture is heated, then the matrix absorbs the excess heat, and the explosion can be avoided. Two decomposition steps can be observed. In the first step (at 65 °C) in addition to the 14.1 % mass decrease, a two-component solid residue forms [reaction 4]. The components of this residue can be easily separated, as **F** is well soluble in water, while **G** does not dissolve at all. The **F:G** mass ratio is 1.00 : 2.97. On further heating, **F** decomposes without a solid residue [reaction 5]. That means pure **G** is the final solid decomposition product of **A**. **G** contains two other elements in addition to 27.0 wt% oxygen. One of the two components of the gas mixture forming in the first decomposition step can be easily quantitated if it is absorbed in acid solution.

Some hints

- Compound **A** decomposes without releasing oxygen.
- Compound **A** contains two different metals.
- Compounds **A** and **B** both contain a complex ion.
- **C** is a well-known compound for any chemistry student.
- Compound **F** does not contain any metal.

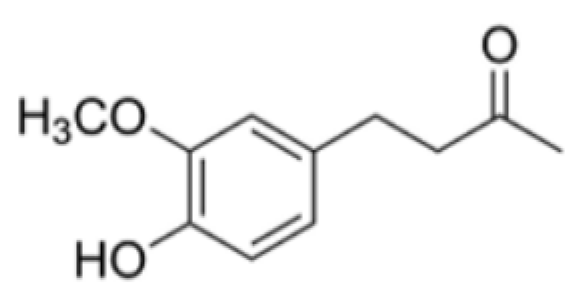
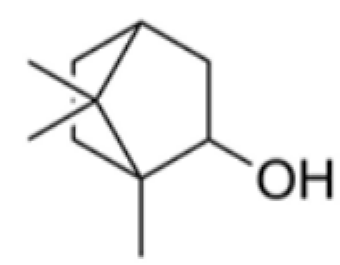
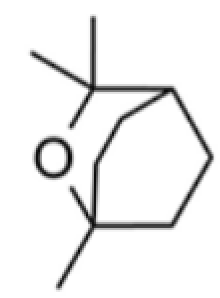
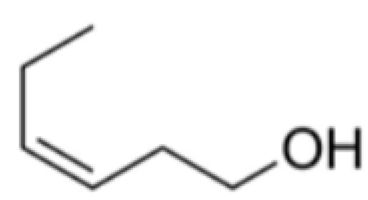
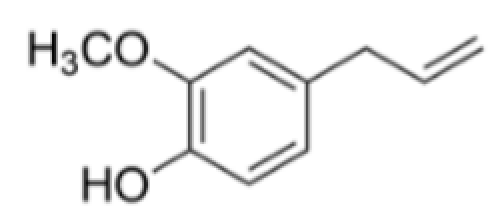
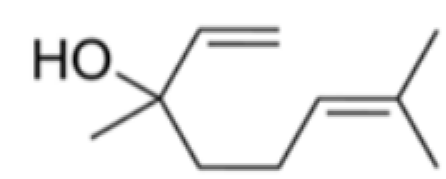
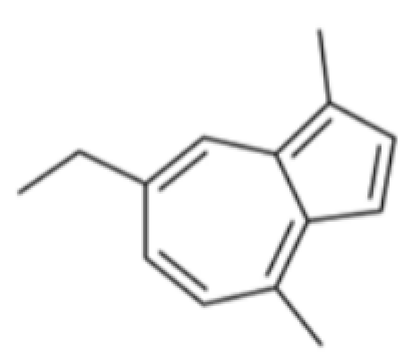
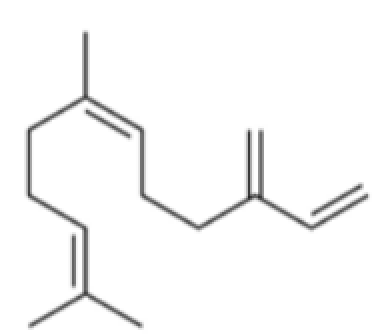
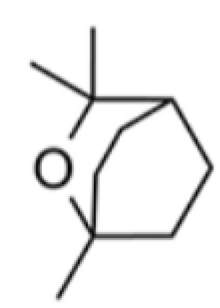
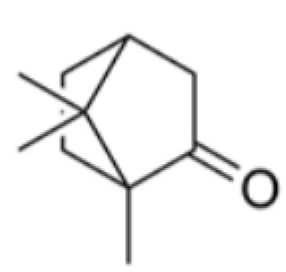
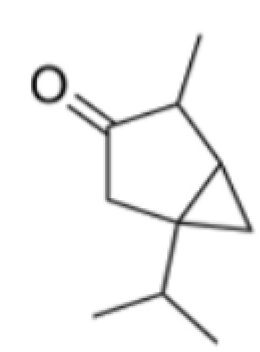
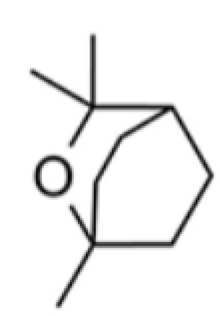
3.2 Give chemical formulae of **A—G**. You don't have to show your calculations, but if your compounds are incorrect you may gain partial marks from correct calculations.

3.3 Write balanced equations for reactions **[3]—[5]**.

A chemist found herself in the 11 th Century, where she met the renowned physician Avicenna. In Avi cenna's laboratory, she discovered a blue elixir and a yellow mysterious stone. She brought them back to 2026. Determined to uncover their composition, she began her research.

Part 1. The blue elixir

At first, the chemist compiled a table listing the names of the only four plants which she saw in Avicenna's lab that could have been used to prepare the elixir, and the compounds that can be extracted from these plants.

Plants	Compounds that can be extracted		
Zingiber			
	1 ($C_{11}H_{14}O_3$)	**2** ($C_{10}H_{18}O$)	**3** ($C_{10}H_{18}O$)
Hypericum			
	4 ($C_6H_{12}O$)	**5** ($C_{10}H_{12}O_2$)	**6** ($C_{10}H_{18}O$)
Chamomilla			
	7 ($C_{14}H_{16}$)	**8** ($C_{15}H_{24}$)	**3** ($C_{10}H_{18}O$)
Artemisia			
	9 ($C_{10}H_{16}O$)	**10** ($C_{10}H_{18}O$)	**3** ($C_{10}H_{18}O$)

Using chromatography, she determined that the elixir consists of four different substances (X, Y, Z and W).

The chemist then discovered that X can isomerise into Y in an acidic medium and that Y has a plane of symmetry.

1.1 Identify X and Y.

Chromatography also revealed it was substance Z that gave the elixir its blue colour. Z is a derivative of the well-known substance A, which has two perpendicular planes of symmetry. When heated, A isomerises into aromatic compound B, which has three mutually perpendicular planes of symmetry. When substance Z is chlorinated, a pair of peaks with an intensity ratio of 3:1 is observed in the mass spectrum at m/z 218 and 220, respectively.

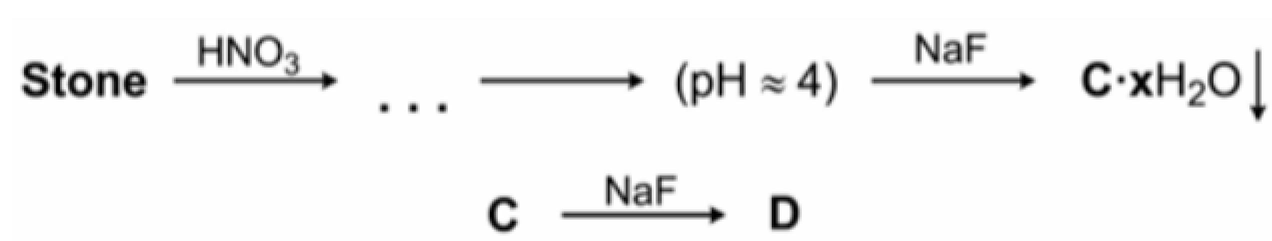
1.2 Identify Z and draw the structures of substances A and B.

The chemist extracted W from the elixir. When W was added to an aqueous Fe^{3+} solution, a characteristic colour change was observed. Mass spectrometric analysis of W revealed the intensity ratio of $[M]^+ : [M + 1]^+ = 9:1$, where M is the molecular ion. Assume carbon consists exclusively of ^{12}C and ^{13}C, the natural isotopic abundance of ^{12}C is 98.9%, and all other elements are monoisotopic.

1.3 Determine the number of carbon atoms (n) from the mass spec data and identify W. Show your calculations.

Part 2. The mysterious stone

Avicenna had told her the stone was a stoichiometric compound. She dissolved it in dilute nitric acid. After that, she adjusted the pH of the solution to ~4 and added NaF. This led to the formation of precipitate $C \cdot xH_2O$, which contained 39.16% water by mass. Anhydrous C reacts with an excess of NaF to yield compound D, which contains 32.85% sodium and 12.85% of metal Q by mass, and is used in the industrial production of Q.



1.4 Identify metal Q and compounds $C \cdot xH_2O$ and D.

Avicenna told her that he found the stone in a coal deposit. The stone contains the anion of a highly symmetrical acid F. Reaction of F with P_2O_5 gives binary compound G, which contains 49.98% of oxygen by mass and has a three-fold symmetry axis. F can be prepared by oxidising compound E with potassium permanganate in an acidic solution. E contains 11.18% of hydrogen by mass and has a six-fold symmetry axis.

1.5 Draw the structures of E, F, and G.

Then, she analysed the stone thermogravimetrically in open air. A 10.00 g sample began to lose mass at approximately 100 °C and stopped at 5.75 g at 200 °C. A second drop in mass was observed at 400 °C, with a final mass of 1.50 g of compound H remaining constant at higher temperatures.

1.6 Determine the chemical formulae of the stone and compound H using thermogravimetric data.